Problem Set 2 - IC104

Solutions

Essential Mathematics for Economic Analysis, 4th edition

Page 118. Question 9

Suppose that all prices rise at the same proportional rate in a country whose inflation rate is 19% per year. For an item that currently costs P_0, use the implied formula:

$P(t) = P_0(1.19)^t$

for the price after t years in order to predict the prices of:

(a) A 20 kg bag of corn, presently costing \$16, after 5 years.

(b) A \$4.40 can of coffee after 10 years.

(c) A \$250,000 house after 4 years.
Answers:

$$a) P(5) = P_0(1.19)^5 = 16 \cdot 1.19^5 \simeq 38.18$$
$$b) P(10) = P_0(1.19)^{10} = 4.4 \cdot 1.19^{10} \simeq 25.06$$
$$c) P(4) = P_0(1.19)^4 = 250,000 \cdot 1.19^4 \simeq 501,334.8$$

Page 123. Question 2

Solve the following equations for x:

(a) $3^x = 8$; (b) $ln(x) = 3$; (c) $ln(x^2 - 4x + 5) = 0$

(d) $ln[x(x-2)] = 0$; (e) $\frac{x \cdot ln(x+3)}{x^2+1} = 0$; (f) $ln(\sqrt{x} - 5) = 0$

Answers:

$$
\begin{aligned}
&a)\ln 3x = \ln 8\\
&\quad x\ln 3 = \ln 8\\
&\quad \Rightarrow x = \frac{\ln 8}{\ln 3}\\
&b)x = e^3 \simeq 20.0855\\
&c)x^2 - 4x + 5 = 1\\
&\quad x^2 - 4x + 4 = 0\\
&\quad \Rightarrow (x-2)^2 = 0 \Rightarrow x = 2\\
&d)x(x-2) = 1\\
&\quad x^2 - 2x - 1 = 0 \Rightarrow x = 1 \pm \sqrt{2}\\
&e)x = 0 \quad or \quad \ln(x+3) = 0 \Rightarrow (x+3) = e^0 = 1 \rightarrow \quad x = -2\\
&f)\sqrt{x} - 5 = 1 \Rightarrow \sqrt{x} = 6 \Rightarrow x = 36
\end{aligned}
$$

Page 162. Question 5 a) b) c) d)

In each case, find the slope of the tangent to the graph of f at the specified point:

(a) $f(x) = 3x + 2$ at (0, 2); (b) $f(x) = x^2 - 1$ at (1,0)

(c) $f(x) = 2 + 3/x$ at (3,3); (d) $f(x) = x^3 - 2x$ at (0,0)

Answers:

$$
\begin{aligned}
&a)f'(x) = 3; \quad \text{the slope is 3.}\\
&b)f'(x) = 2x|_{x=1} = 2; \quad \text{the slope is 2.}\\
&c)f'(x) = -3x^{-2}|_{x=3} = -\frac{1}{3}; \quad \text{the slope is } -\frac{1}{3}.\\
&d)f'(x) = 3x^2 - 2|_{x=0} = -2; \quad \text{the slope is } -2.
\end{aligned}
$$

Page 177. Question 3

Find the derivatives of the following:

(a) x^6 ; (b) $3x^{11}$; (c) x^{50} ; (d) $-4x^{-7}$

(e) $\frac{x^{12}}{12}$; (f) $-\frac{2}{x^2}$; (g) $\frac{3}{x^{\frac{1}{3}}}$; (h) $-\frac{2}{x\sqrt{(x)}}$

Answers:
a) $6x^5$; b) $33x^{10}$; (c) $50x^{49}$; (d) $28x^{-8}$

(e) x^{11} ; (f) $4x^{-3}$; (g) $-x^{-\frac{4}{3}}$; (h) $3x^{-\frac{5}{2}}$

Page 183. Question 2 b) c)

Differentiate:

(b) $(2x^2-1)(x^4-1)$; c) $(x^5+\frac{1}{x})(x^5+1)$

Answers:

$$b)4x(3x^4-x^2-1)$$
$$c)10x^9+5x^4+4x^3-x^{-2}$$

Differentiate:

Page 183. Question 3 d) h)

(d) $\frac{x+1}{x-1}$; h) $\frac{3x-1}{x^2+x+1}$

Answers:

$$d)f'=\frac{(x-1)-(x+1)}{(x-1)^2}=\frac{-2}{(x-1)^2}$$
$$h)f'=\frac{3(x^2+x+1)-(3x-1)(2x+1)}{(x^2+x+1)^2}=\frac{-3x^2+2x+4}{x^2+x+1^2}$$

Page 187. Question 2

Compute the following:

(a) $\frac{dY}{dt}$ when $Y=-3(\mathrm{V}+1)^5$ and $V=\frac{1}{3}t^3$

(b) $\frac{dK}{dt}$ when $K=Al^a$ and $L=bt+c$ (A, a , b and c are positive constants.)

Answers:

$$a)\frac{dY}{dt} = (\frac{dY}{dV})(\frac{dV}{dt}) = (-3)5(V+1)^4 t^2 = -15t^2(\frac{t^3}{3}+1)^4$$
$$b)\frac{dK}{dt} = (\frac{dK}{dL})(\frac{dL}{dt}) = AaL^{a-1}b = Aab(bt+c)^{a-1}$$

Page 193. Question 1

Compute the second derivatives of:

(a) $y = x^5 - 3x^4 + 2$; (b)$y = \sqrt{x}$; (c) $y = (1+x^2)^{10}$

Answers:

$a)y' = 5x^4 - 12x^3, \quad y'' = 20x^3 - 36x^2$
$b)y' = \frac{1}{2}x^{-\frac{1}{2}}, \quad y'' = \frac{1}{2}(-\frac{1}{2})x^{-\frac{3}{2}} = -\frac{1}{4}x^{-\frac{3}{2}}$
$c)y' = 20(1+x^2)^9 x, \quad y'' = 180(1+x^2)^8 2xx + 20(1+x^2)^9 = 360(1+x^2)^8 x^2 + 20(1+x^2)^9$
$=20(1+x^2)^8(1+19x^2)$

Page 197. Question 3 (a) (b)

Find the first and second-order derivatives of:

(a) $y = e^{-3x}$; (b)$y = 2e^{x^3}$

Answers:

$a)y' = -3e^{-3x}, \quad y'' = 9e^{-3x}$
$b)y' = 2e^{x^3}3x^2 = 6e^{x^3}x^2, \quad y'' = 6e^{x^3}3x^2x^2 + 6e^{x^3}2x = 18e^{x^3}x^4 + 12e^{x^3}x$
$= 6xe^{x^3}(3x^3+2)$